KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF ORTHOPAEDIC AND TRAUMA MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN ORTHOPAEDIC AND TRAUMA MEDICINE

PAPER: PRINCIPLES OF ORTHOPAEDICS

DATE: 4TH FEBRUARY 2025 **TIME: 3 HOURS (9:00 – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answers Question)
2. Attempt **ALL** Questions
3. **ALL** Multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write your **EXAMINATION NUMBER** on all the answer sheets provided and on the question paper.
5. No other writing should be done on the question paper except your Examination Number.
6. Ensure that all examination answer scripts are handed in at the end of the examination
7. Ensure you sign the examination register provided

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. A patient present with history of night sweats, weight loss and a deformity on his back. What is your diagnosis?
   A. Potts disease
   B. Septic arthritis
   C. Chronic osteomyelitis
   D. Leg-calve Perthe' disease
2. Which one of the following is an inflammatory arthritis?
   A. Rheumatoid arthritis
   B. Osteoarthritis
   C. Osteomyelitis
   D. Myositis ossificans
3. Genu varum is also known as:
   A. bow legs
   B. knocked knees
   C. pes-cavus
   D. genu recurvatum
4. Early CTEV is treated by?
   A. Casting
   B. Traction
   C. operation repair
   D. strapping
5. The term orthopedic was first coined by
   A. Robert jones
   B. Nicholas Andre
   C. Nicholas Adrian
   D. Owen Thomas
6. During embryological development, the nervous system is derived from which germ layer.
   A. Ectoderm
   B. Mesoderm
   C. Endoderm
   D. Epiderm
7. The definitive treatment of chronic osteomyelitis includes:
   A. Sequestrectomy
   B. Analgesics and antipyretics
   C. Limb elevation
   D. Corticosteroids
8. A patient has been diagnosed to have acute osteomyelitis, which is most likely causative organism
   A. streptococcus haemolyticus
   B. clostridium welchii
   C. Escherichia coli
   D. staphylococcus aureus

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9. Which one of the following is a non-steroidal anti-inflammatory drug for rheumatoid arthritis treatment?
   A. Tramadol
   B. Meloxicam
   C. Pethidine
   D. Methotrexate
10. Bone sequestrum is commonly associated with:
   A. Tuberculous osteomyelitis
   B. Syphilitic osteoperiostitis
   C. Acute osteomyelitis
   D. Chronic osteomyelitis
11. bony ankylosis is caused by:
   A. Rheumatoid arthritis
   B. Osteoarthtis
   C. Septic arthritis
   D. Gouty arthritis
12. Urate crystal deposits in joints are pathognomonic of which condition?
   A. Pseudogout arthritis
   B. Gouty arthritis
   C. Psoriatic arthritis
   D. Osteoarthritis
13. The most common site for acute osteomyelitis in children is
   A. Radius
   B. Tibia
   C. Femur
   D. Humerus
14. Onion peel appearance is characteristic of which one of the following tumours?
   A. Ewings sarcoma
   B. Giant cell tumour
   C. Osteogenic sarcoma
   A. Osteochondroma
15. A 25-year-old female patient with multiple peripheral joint pains with early morning stiffness. He also has generalized body weakness and raised ESR. This patient has:
   A. Osteoarthritis
   B. Ankylosing spondylitis
   C. Rheumatoid arthritis
   D. Gout arthritis
16. Which one of the following positions of the interphalangeal joints characterize boutonniere deformity of the index finger?
   A. Proximal and distal interphalangeal flexion
   B. Distal interphalangeal flexion proximal extension
   C. Proximal interphalangeal flexion and distal extension
   D. Proximal and distal interphalangeal extension

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17. Paget's disease is also known as?
   A. Osteitis fibrosa cystica
   B. Osteoporosis
   C. Osteitis deformans
   D. Osteopetrosis
18. Which one of the following is a typical characteristic of pes planus?
   A. High arched foot
   B. inverted foot
   C. Everted foot
   D. Flat foot
19. Which one of the following is associated with tennis elbow?
   A. Tenderness at the cubital fossa
   B. Tenderness on the olecranon process
   C. Tenderness on the medial epicondyle
   D. Tenderness on the lateral epicondyle
20. Barlows test is done for
   A. Dislocation if patella
   B. Perthes disease
   C. Congenital hip dislocation
   D. Shoulder dislocation
21. A 10-year-old girl has fever and swelling in the lower part of the left leg for two days. There is intense tenderness over the swelling x-ray done appears normal. What is the likely diagnosis?
   A. Septic arthritis
   B. rheumatoid arthritis
   C. Acute osteomyelitis
   D. Gouty arthritis
22. Which one of the following is a sign of rheumatoid arthritis?
   A. Swan neck deformity
   B. Morning stiffness
   C. Multiple joint pains
   D. Joint swelling
23. Which one of the following is an important test in the diagnosis of osteoarthritis?
   A. Serum uric acid levels
   B. Anti-mutate citrullinated vimetin (anti-mcv)
   C. Rheumatoid factor
   D. Plain X-ray of the joint
24. Which one of the following is a common site for osteogenic sarcoma?
   A. Pelvis
   B. Distal tibia
   C. Distal humerus
   D. Distal femur

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25. Which one of the following is true about osteochondroma?
    A. It grows towards the joint
    B. The tumour is capped with cartilage
    C. It is a malignant tumour
    D. It is a tumour of the skull
26. The gold standard for diagnosis of nerve compression in the spine is:
    A. CT scan of the spine
    B. Magnetic resonance imaging
    C. CT scan guided biopsy for acid fast bacilli
    D. Plain X-ray of the spine
27. The organism that has a special predilection in many cases of osteomyelitis in sicklers is:
    A. Kingeila kingie
    B. Salmonella typhi
    C. Escheria coil
    D. Pseudomonas aeruginosa
28. Which one of the following is a sign of rickets?
    A. Bone pains
    B. Generalized weakness
    C. Rachitic rosary
    D. Headache
29. 10-year old patient presents with bowing of legs which could be attributed to vitamin D deficiency. What is the likely diagnosis in this patient?
    A. Rickets
    B. Osteomalacia
    C. Osteoporosis
    D. Scurvy
30. A positive Ortolani test is one that:
    A. Dislocates a dislocatable shoulder
    B. Dislocates a dislocatable hip
    C. Reduces a dislocated hip
    D. Reduces a dislocated shoulder
31. A patient presents with popliteal pain, worsened by activity. On examination, there is a localized fluctuant swelling in the popliteal area. This is likely to be:
    A. Myositis ossificans
    B. Pre-patellar bursitis
    C. Baker's cyst
    D. Torn meniscus
32. Which one of the following is a modifiable risk factor for osteoarthritis?
    A. Age
    B. Gender
    C. Large body mass
    D. Genetics

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33. Which one of the following ligaments prevents excessive adduction of the knee joint?
   A. Lateral collateral ligament
   B. Anterior cruciate ligament
   C. Medial collateral ligament
   D. Posterior cruciate ligament
34. Carpal tunnel syndrome is a condition that affects the hand and is caused by pressure to:
   A. Ulnar nerve
   B. Radial artery
   C. Median nerve
   D. Radial nerve
35. When correcting clubfoot using Ponseti method, the goal is to:
   A. Rotate foot medically around a fixed talus
   B. Rotate foot laterally around a fixed talus
   C. Rotate foot laterally around a fixed navicular
   D. Rotate foot laterally around a fixed calcaneus
36. An adolescent present with a painful swelling around the knee joint especially at night. radiograph shows an ill-defined mass with sun burst appearances. What is the investigation of choice?
   A. plain X-ray
   B. bone biopsy
   C. MRI
   D. Bone CT scan
37. An adult patient presents with pain in the knee medially for a few weeks. Radiograph shows a detached fragment of well circumscribed bone on the medial condyle. The diagnosis is:
   A. Torn Meniscus
   B. Osteochondritis dissecans
   C. Tear of anterior cruciate ligament
   D. Osteoarthritis of the knee
38. What is the earliest bony changes of osteomyelitis on X-rays?
   A. Sequestrum
   B. Lytic defects
   C. Loss of soft tissue plane
   D. Periosteal reaction
39. The deformity that is corrected first in clubfoot is:
   A. Forefoot abduction
   B. Forefoot adduction
   C. Inversion of the foot
   D. Equinus deformity
40. A 15year old child with history of fall develops painful knee joint swelling. On examination, the joint is warm, fluctuant and tender. Which curative drug will you use for this patient?
   A. Indomethacin
   B. Prednisolone
   C. Tramadol
   D. Flucloxacillin

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SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. A female patient presents with morning stiffness of the small joints of the feet and hands, the ESR is elevated. Diagnose and outline the pathophysiology of this condition **(5 marks)**
42. A 10-year-old child present with deformity of the ankle joint. Highlight five (5) possible causes of this deformity **(5 marks)**
43. Compare and contrast benign and malignant bone tumours **(5 marks)**
44. Outline five (5) signs of a child with nutritional rickets **(5 marks)**
45. A 20-year-old patient presenting with pain and swelling on the proximal part of the leg for four months. On examination you recognize features of chronic osteomyelitis. Highlight anticipated signs on local examination and radiological features **(5 marks)**
46. Explain how you would grade muscle power for a patient who present outpatient orthopaedic clinic with weakness of the left arm **(5 marks)**
47. A patient presents to the orthopaedic clinic limping. You realize that he has limb length discrepancy. Explain the procedure of measuring lower limb length **(5 marks)**
48. Highlight the clinical and radiological features in a patient with persistent congenital hip dysplasia **(5 marks)**

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49. Outline the management of a neonate with club foot **(20 marks)**

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